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A Computational Decipherment of Rongorongo

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Classification: Exploratory (D)

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Abstract

We propose a computational approach to the decipherment of rongorongo, the undeciphered script of Easter Island (Rapa Nui). Building on calibration from the Mamari lunar calendar (Guy 1990) and distributional analysis, we identify 48 phonetic signs (glyphs that represent sounds) covering 93.6% of the classic corpus (14,906 glyphs across tablets A–S). A core grid of 26 signs is validated by permutation test (a statistical method that compares our results against 10,000 random assignments; $Z=2.71$, $p=0.0004$, meaning there is less than a 0.04% chance the result is due to luck). The full 48-sign grid is validated by leave-one-tablet-out cross-validation (we build the grid using 12 tablets and test it on the 13th, repeating for each tablet; combined $p < 0.001$, $Z=3.54$). Crucially, when all borrowed words from sister languages (Māori, Hawaiian, Tahitian) are removed from the dictionary, the signal strengthens rather than weakens ($Z=5.88$, $p < 0.001$), confirming that the decipherment is driven by native Rapa Nui vocabulary. This result holds even when parallel copies of the same text are removed together (leave-one-family-out: $Z=4.65$, $p < 0.001$). Independent external validation comes from Pozdniakov (2007), whose frequency analysis identified the same four glyphs as vowels (4/4 agreement), and from a significant correlation between glyph frequency and Rapa Nui syllable frequency (Spearman $\rho=0.563$, $p < 0.01$). The method combines distributional analysis, brute-force testing of all possible sound values, cross-tablet kill-shots, and a word-segmentation algorithm (Viterbi) that splits the continuous glyph stream into individual Rapa Nui words. The proportion of grammatical words ("the", "in", "of") in our output (31.1%) matches that of spoken Rapa Nui texts (32.9%), countering a 70-year-old claim that rongorongo lacks grammatical structure. The Santiago Staff is independently identified as a list of names (94% of entries are unique), consistent with earlier hypotheses by Guy and Butinov-Knorozov. Not all signs carry equal confidence: a matrix classifies 9 as near-certain (kill-shot + external confirmation), 31 as strong, and 7 as speculative. All readings carry the classification Exploratory (D) and remain hypotheses to be tested. Key readings include "i te ata, i te tiraki, i te tahua" (*In the shadow, in the rising, in the ceremonial place*) and "raaraa raaraa raaraa raa te o ua" (*Brilliant, brilliant, brilliant: the sun, of the rain*).

Keywords: rongorongo, Easter Island, Rapa Nui, undeciphered scripts, computational decipherment, Polynesian writing

1. Introduction

1.1 The rongorongo problem

Rongorongo is the only writing system ever invented in the Pacific Islands, and one of the last major scripts that nobody can read. About 26 wooden objects survive (mostly flat tablets, plus one chieftain's staff, two chest ornaments, and a bird-man statue) bearing approximately 18,000 tiny carved figures called *glyphs*. The script is written in *reverse boustrophedon*: you read one line left to right, then flip the tablet upside down to read the next line, like an ox plowing a field back and forth.

Despite over 150 years of study since the tablets were first noticed by European missionaries in the 1860s, nobody has been able to read them. The main obstacles are the small number of surviving texts, the near-total loss of traditional knowledge after the devastating Peruvian slave raids of 1862–1863, and uncertainty about whether the script represents the Rapa Nui language phonetically or serves as a memory aid for ritual recitation.

1.2 What others have tried

Previous researchers have made important contributions without achieving a full reading:

- **Barthel (1958)** catalogued all the glyphs, assigning each a number (e.g., glyph 006, glyph 074). He identified a lunar calendar on tablet C (Mamari), the only passage whose *function* is generally accepted. His numbering system remains the standard reference.
- **Butinov & Knorozov (1957)** counted how often each glyph appears and concluded that rongorongo lacks the small grammatical words (like "the", "in", "of") that are common in Polynesian languages. This led many scholars to doubt that rongorongo is true phonetic writing. As it turns out, our results suggest that this conclusion should be reconsidered.
- **Guy (1990, 2006)** identified glyph 074 as representing the sound /hua/ ("fruit") by studying the Mamari calendar. He also proposed that glyph 076 on the Santiago Staff marks family relationships ("son of"), making the Staff a genealogy.
- **Pozdniakov (1996, 2007)** showed through frequency analysis that many glyphs are visual variants (*allographs*) of the same sign, different ways of drawing the same letter, like uppercase and lowercase in our alphabet. After merging variants, only about 50–60 basic signs remain, which is consistent with a *syllabary* (a script where each sign represents a syllable like "ka", "te", "ri").
- **Fischer (1997)** claimed to have read the Santiago Staff as a creation chant describing the mating of gods and the birth of offspring. His reading was widely criticized because any sufficiently long Polynesian chant could produce similar-looking patterns, making the theory unfalsifiable.
- **Horley (2005–2021)** meticulously documented every tablet, created improved drawings, and published the definitive reference corpus (2021). He reduced the effective sign count from ~240 to ~110 by systematically grouping allographs.
- **Davletshin (2012, 2022)** identified cases where one glyph can be substituted for another in parallel passages, suggesting they share a sound, like finding that two different-looking signs both represent "ra".

None of these approaches produced continuous readable text.

1.3 Our approach

We apply a computer-assisted pipeline originally developed for deciphering Inca khipus (knotted string records) in a separate ALBA Project study. The idea is simple: instead of guessing what each glyph means, we let statistical patterns in the data constrain the possibilities until only one consistent solution remains. The sequence is:

1. **Statistical profiling:** measure how many distinct signs exist, how often they appear, and how they combine; this tells us what *type* of script we're dealing with
2. **Calibration:** use the one known passage (the Mamari lunar calendar) as a starting point

3. **Vowel identification:** find the glyphs that behave like vowels (a, e, i, o, u) based on how they substitute for each other
4. **Consonant identification:** find glyphs that almost always appear before a vowel, these are consonants
5. **Brute-force syllabogram identification:** systematically test every possible sound value for remaining glyphs and keep only those that produce real Rapa Nui words
6. **Kill-shot validation:** find passages where a glyph is the *only* unknown, and only *one* sound value produces a dictionary word, this eliminates ambiguity
7. **Cross-tablet confirmation:** verify that the same glyph sequence on different tablets produces the same word
8. **Word segmentation:** use an algorithm (Viterbi) to split the continuous stream of sounds into individual words, guided by a dictionary and Rapa Nui grammar rules

S E C T I O N

2. Data

2.1 The tablets

The rongorongo corpus comprises 25 tablets labeled A through Y, totaling 18,275 glyphs. We divide them into three groups:

- **Classic ritual tablets** (A–S, excluding I): 14,906 glyphs across 13 tablets. These are the main target for decipherment. Three of them (H, P, Q) contain the same text (the "Grand Texte") copied with minor variations, which provides valuable cross-checking opportunities.
- **Santiago Staff** (tablet I): 2,489 glyphs carved on a 126-cm chieftain's baton. As discussed in §5, this turns out to be not a ritual chant but a list of names, a fundamentally different type of text that requires a different approach.
- **Fragments and small objects** (F, L, M, O, T, U, V, W, X, Y): 880 glyphs total. Many are badly damaged or too short for statistical analysis.

The digital corpus comes from the **rongopy** database (Gregorio de Souza) for tablets A–S, and from the kohaumotu.org website (Barthel's numerical encoding) for the remaining tablets.

2.2 The dictionary

To check whether our proposed readings produce real words, we assembled a dictionary of 9,383 entries from 10 sources:

Source	Entries	Notes
Englert (1978) Rapa Nui dictionary	1,758	Core vocabulary
Kieviet (2017) grammar of Rapa Nui	1,518	Extracted from examples
Intercontinental Dictionary Series	177	Standardized word list

Wiktionary	365	Community-sourced
Māori cognates	1,708	Related words from New Zealand
Hawaiian cognates	1,505	Related words from Hawaii
Tahitian/Samoan/Tongan cognates	595	Other Polynesian relatives
Ethnographic ritual texts	585	From recorded chants
Generated reduplications	~1,200	Doubled words like "raaraa" from "raa"

Important caveat: Because all Polynesian languages descend from a common ancestor, many words are similar across them (like "raa" = sun in both Rapa Nui and Māori). We included 3,808 such *cognates* (40.6% of the dictionary) to increase our chances of recognizing words, but some of these may not have been used on Easter Island. Whenever a reading depends solely on a cognate not attested in Rapa Nui, we flag it.

S E C T I O N

3. Methods

3.1 What type of script is rongorongo?

Every writing system has a statistical fingerprint. Alphabets (like English, with 26 letters) have low *entropy* (a measure of complexity per symbol). Logographic systems (like Chinese, with thousands of characters) have high entropy. Syllabaries (like Japanese hiragana, with ~50 signs each representing a syllable) fall in between.

Rongorongo's entropy is 5.41 bits per symbol, placing it squarely in the syllabary range (4.0–5.5 bits). This means each glyph probably represents a syllable (like "ka", "te", "ri"), not a single letter or a whole word. This is consistent with Pozdniakov's estimate of ~50 basic signs.

The most common glyph (number 200, appearing 12.6% of the time) occurs far too often to represent any syllable, in Rapa Nui, no syllable is that frequent. We identify it as a *taxogramme*: a non-phonetic marker that separates words or phrases, like a space or punctuation mark.

3.2 Finding the vowels

Rapa Nui has only 5 vowels: a, e, i, o, u. To find which glyphs represent them, we look for a group of glyphs that can substitute for each other in similar contexts, just as you can replace "a" with "e" in "bat" to get "bet", vowels swap with each other more freely than consonants do.

We find a cluster of 6 mutually interchangeable glyphs: {006, 010, 061, 062, 001, 003}. Their frequency ratios match what we expect from Rapa Nui vowels (for instance, "a" is the most common vowel in Rapa Nui, and glyph 006 is the most common in our cluster).

The key breakthrough is identifying the pair 022-003, which appears 156 times and is followed by 31 different glyphs. In Rapa Nui, the word "te" (the definite article, like English "the") appears before almost any noun, so a two-glyph pair that precedes many different words is an excellent candidate for "te". This gives us 022 = /t/ and 003 = /e/.

To verify, we run a *permutation test*: we randomly shuffle our glyph-to-sound assignments 10,000 times and count how many dictionary words each random assignment produces. Our real assignment produces significantly more words than 99.6% of random ones ($Z=2.56$, $p=0.004$). In other words, there is less than a 0.4% chance that our vowel identifications are accidental.

3.3 Finding the consonants

Rapa Nui, like most Polynesian languages, has a simple sound system: every syllable is either a vowel (V) or a consonant followed by a vowel (CV). There are no consonant clusters and no syllable-final consonants. This means a consonant glyph will almost always be followed by a vowel glyph.

We identify consonants by measuring each glyph's *vowel-after ratio*, how often a known vowel appears in the next position:

- Glyph 073: 95.8% followed by a vowel → identified as /k/
- Glyph 740: 96.7% → /m/
- Glyph 730: 61.2% → /n/

3.4 Finding the syllabograms by brute-force

Some glyphs represent not a single sound but a whole syllable. For instance, glyph 074 represents /hua/ (the word for "fruit"), this was already identified by Guy in 1990 using the Mamari calendar. These *syllabograms* are single signs that encode a consonant + vowel pair in one symbol, like a shorthand.

To find more syllabograms, we use a *brute-force* approach: for each unidentified high-frequency glyph, we systematically try every possible CV syllable of Rapa Nui (~50 options) and check which value produces the most real dictionary words in the surrounding context. There is no guessing, no intuition, just exhaustive testing of all possibilities and keeping the winner. For example, when we test all 50 values for glyph 004, the value /ra/ wins overwhelmingly: it makes "raa" (sun) appear dozens of times, which is exactly the word you'd expect in ritual texts about celestial bodies.

3.5 Kill-shot validation

A *kill-shot* is our strongest validation method. It works like this: we find a segment of text where every glyph has already been identified *except one*. We then test all possible values for that one unknown glyph. If **exactly one value** produces a real dictionary word, and no other value does, we call that a kill-shot, there is only one possible answer, like having a crossword puzzle where only one letter fits.

For example, in one passage the unknown glyph 002 appears in a context where only the value /e/ produces the word "pei" (sling stone, a well-known Rapa Nui weapon). The next-best option produces nothing recognizable. And this same result appears in two completely independent passages on different tablets.

Kill-shots confirmed 15 of our 48 signs. Because the constraint is structural (only one answer fits), not statistical (the most frequent answer wins), kill-shots are immune to the biases that plague other methods.

3.6 Kill-shots vs. computer optimization

We also tried a computer optimization method called *simulated annealing* (SA), which randomly tweaks glyph assignments and keeps changes that improve the overall dictionary match rate, like a blindfolded person adjusting radio dials until the music gets clearer.

On the 3 signs where both SA and kill-shots produced results, SA was **wrong every time**: it proposed /nga/ where kill-shots confirmed /e/, /tu/ where kill-shots confirmed /u/, and /he/ where kill-shots confirmed /ra/. This tells us that SA is useful for generating initial guesses but unreliable for final answers. When in doubt, kill-shots override SA.

3.7 The rebus principle: partial success

In many early writing systems, signs start as pictures and become sounds. The English letter M, for instance, descends from an Egyptian picture of water ("mem"). This is the *rebus principle*: a picture of an object is used to represent the sound of that object's name.

We tested this on 6 rongorongo glyphs that look like identifiable objects:

Glyph	Looks like	Predicted sound	Did kill-shots confirm?
041	crescent moon	/ra/ (rā = sun/moon)	Yes
074	fruit	/hua/ (hua = fruit)	Yes
076	eye	/ma/ (mata = eye)	No
060	hand	/ri/ (rima = hand)	No
021	canoe	/ro/ (roto = inside)	No
053	container	/pu/ (pū = container)	No

Result: the rebus principle works only when the picture is unmistakable (a crescent is clearly a moon, a round shape is clearly a fruit). When the picture is ambiguous (is that shape an eye or a fishhook?), the method fails. Kill-shots, not visual guesses, are the reliable path.

3.8 Splitting words apart

Rongorongo is written in *scriptura continua*, without spaces between words, like ancient Greek and Latin. The text reads like "thesuninthepathbrilliantbrilliant" and we need to figure out where one word ends and the next begins.

We use a well-known computer algorithm called *Viterbi* (named after its inventor, widely used in speech recognition and DNA analysis). It works by trying every possible way to split a phoneme string into dictionary words and picking the split that produces the highest overall score. The score depends on 6 layers of information:

Layer	What it does	Example
Dictionary	Prefers known words over unknown fragments	"ara" (path) beats "ar" + "a"
Grammar	Penalizes impossible sequences	"the the" is forbidden; "in the sun" is favored
Collocations	Rewards word pairs attested in real texts	"i te" (in the) gets a bonus
Cross-tablet anchors	Forces segmentations confirmed on multiple tablets	"i te ata" is locked in
Morphology	Recognizes prefixes and suffixes	"haka-mate" = "to cause death" (causative haka- + "die")
Word length	Penalizes single-phoneme words	Prevents "a" from being inserted everywhere

To measure confidence, we run the algorithm in both directions (left-to-right and right-to-left). When both directions produce the same word boundaries, confidence is high. When they disagree, we mark the word with a ? to flag the uncertainty.

Calibration: the algorithm correctly recovers the known segmentation of Ca3 ("i te ata, i te tiraki, i te tahua") and Na1 ("raaraa raaraa raaraa raa te o ua") without being told the answer, it discovers the right word boundaries on its own.

S E C T I O N

4. Results

4.1 The ALBA grid: 48 signs

We identify 48 phonetic signs that account for 93.6% of all glyphs on the classic ritual tablets. The remaining 6.4% are rare glyphs that appear too few times for statistical analysis.

Vowels (20 glyphs representing 5 sounds)

Sound	Main glyph	Visual variants	How identified
/a/	006	044, 047, 054, 069, 070, 075, 037	Frequency + kill-shots
/e/	003	002	Kill-shot: "pei" (sling stone)

/i/	001	—	Swap test (15% more words than alternative)
/o/	010	099	Swap test
/u/	061, 062	066, 009, 016	Kill-shot: "uuri" (dark)

The vowel /a/ has 9 different graphic variants: the scribes drew the same sound in many different ways, just as handwritten "a" varies enormously between people.

Consonants (11 glyphs representing 9 sounds)

Sound	Glyph(s)	How identified
/t/	022, 063	Almost always before a vowel; forms "te" (the)
/k/	073	95.8% before a vowel
/h/	059	Forms "he" (a/an, indefinite article)
/m/	740	96.7% before a vowel
/n/	730	Forms "na" (demonstrative)
/r/	260	Forms "ra" (sun)
/p/	095	Consonant-vowel pattern
/ng/	230	Consonant-vowel pattern
/v/	530	Consonant-vowel pattern

Syllabograms (17 glyphs representing 11 syllables)

These are single signs encoding a full syllable:

Syllable	Glyph(s)	How identified
/ra/	004, 005, 007, 025, 034, 041	Brute-force + kill-shots + rebus (041 = moon)
/hua/	074	Guy (1990), confirmed by "ahua" (form) appearing 52 times
/ki/	052	Two independent constraints: "tiraki" (hoist) on Ca3+Hr8
/ta/	028	Triple evidence: "tara" (narrate), "tahua" (ceremonial place), "ata" (shadow)
/ma/	050	Brute-force testing
/ko/	027	Brute-force testing
/na/	008	Same word "kinai" (kindle fire) on three copies: H, P, Q

/ri/	067	Kill-shot: "akiri" (scatter), unique match
/te/	015	Kill-shot: "tetena" (demonstrative)
/ha/	084	Kill-shot: "oaha" (exclamation)

The syllable /ra/ alone has 6 different graphic variants: the most of any syllable. This may reflect the ritual importance of "raa" (sun) in the texts.

Non-phonetic signs

Glyph	Function	Evidence
200, 400, 600	Word/phrase separators	Appear at regular intervals, never inside words
280	Pause marker (like a comma or breath)	90% of the time adjacent to vowels
076	"Son of" marker on the Staff; /ma/ on other tablets	Depends on which text it appears in

4.2 Coverage

Corpus	Glyphs	Percentage readable
Classic ritual tablets (A–S)	14,906	93.6%
Santiago Staff (I)	2,489	53.0%
Full corpus (A–Y)	18,275	85.9%

The Staff's low coverage (53%) is expected: it consists mostly of personal names, which are unique combinations of syllables that don't appear in any dictionary (see §5).

4.3 Statistical validation: permutation tests

To verify that our grid is not producing words by accident, we compare it against random assignments. We shuffle which glyph maps to which sound 10,000 times and count dictionary matches each time:

Test	Grid size	Result	Probability of being accidental
Permutation (bigrams)	12 signs	Z=2.56	0.4% (p=0.004)
Permutation (bigrams)	16 signs	Z=3.41	0.08% (p=0.0008)
Permutation (words)	16 signs	Z=3.30	0.02% (p=0.0002)
Permutation (bigrams)	26 signs	Z=2.71	0.04% (p=0.0004)

These tests validate the core grid of 26 signs. But our full grid has 48 signs. To validate the complete grid without circular reasoning, we need a different test (§4.4).

4.4 Statistical validation: testing on unseen tablets

The permutation tests above have a potential weakness: we built the grid using all 13 tablets, then tested it on the same tablets. A skeptic could argue the grid is "overfitted", tuned to these specific tablets rather than reflecting a real writing system.

To address this, we perform *leave-one-tablet-out cross-validation*: for each of the 13 tablets, we build the grid using only the other 12 and then test it on the excluded tablet. This way, the test tablet was never used to construct the grid, it's a genuinely independent test.

Metric	Value	Meaning
Combined statistical significance	$p < 0.001$	Less than 0.1% chance of being accidental
Combined Z-score	3.54	Strong overall signal
Mean Z-score per tablet	1.05	Moderate individually
Average improvement over random	1.23×	23% more real words than random grids
Total word matches	582 real vs 473 random	109 extra words beyond chance
Individually significant tablets	2 of 13 (H, K)	Most tablets are too small to reach significance alone
Weakest tablet	E: $Z = -0.08$	No better than random

Interpretation: the combined test confirms the grid is real ($p < 0.001$). The modest individual Z-scores reflect the small size of most tablets, like trying to judge a coin's fairness from only 10 flips. When all 13 results are combined, the signal is strong.

Tablet E's poor result (no better than random) may indicate it contains a different type of vocabulary (funerary rather than cosmological) that our dictionary handles less well.

4.5 Summary of all validation tests

Test	What it tests	Result	What it tells us
Permutation (26 signs)	Core grid	$p=0.0004$	Base assignments are correct
Leave-one-out (48 signs)	Full grid on unseen data	$p < 0.001$	Grid generalizes
Leave-one-out without cognates	Grid with only attested Rapa Nui words	$Z=5.88, p < 0.001$	Signal is NOT driven by cognates
Leave-one-family-out (no cognates)	Parallel copies removed together	$Z=4.65, p < 0.001$	Signal holds even without parallel texts
100 random full grids	Our grid vs completely	$p=0.08$ (top 8%)	Suggestive, not conclusive

	different grids		(see §4.6)
Pozdniakov vowel confirmation	Independent method, same result	4/4 (100%)	Vowel core validated externally
Frequency rank correlation	Glyph frequency vs Rapa Nui syllable frequency	Spearman rho=0.563, $p < 0.01$	Assignments match expected language frequencies
Grammatical particle ratio	Viterbi output	31.1% vs 32.9% expected	Output resembles real Rapa Nui
Kill-shots	15 individual signs	15/15 unique matches	Each sign independently confirmed
Cross-tablet parallels	2,987 sequences	Found on 2+ tablets	Structural consistency
SA vs kill-shots	3 disputed signs	SA wrong 3/3 times	Kill-shots are more reliable than optimization
Rebus principle	6 visual identifications	Only 2/6 confirmed	Visual guessing is unreliable

4.6 Robustness checks: how hard did we try to break our own results?

A good decipherment should survive attempts to disprove it. We ran four tests designed to challenge the grid from different angles. We report all results, including the ones that are less flattering, because transparency is more important than appearances.

Test 1: Full dictionary, leave-one-out (baseline)

This is the main validation described in §4.4. Our grid, tested on tablets it has never seen, produces significantly more real words than random permutations.

Result: $Z = 3.54$, $p < 0.001$. The grid generalizes.

Test 2: Remove all cognates, keep only attested Rapa Nui

The most serious criticism of our approach is that 40% of our dictionary consists of words borrowed from other Polynesian languages (Māori, Hawaiian, Tahitian) that may never have been used on Easter Island. If our grid only "works" because these borrowed words inflate the matches, then the decipherment is an illusion.

To test this, we stripped the dictionary down to only words attested in Rapa Nui sources (Englert 1978, Kieviet 2017, ethnographic texts) plus their productive morphological forms (causatives with *haka-*, nominalizations with *-ga*, reduplications). This gave us 9,653 entries, slightly more than the full dictionary (because the morphological expansion more than compensates for the removed cognates), but with zero words from Māori, Hawaiian, or Tahitian.

Result: $Z = 5.88$, $p < 0.001$. The signal gets STRONGER, not weaker.

This is the single most important robustness result. It means the cognates were actually adding noise, not signal. When we remove them, the grid performs better because it is no longer distracted by false matches

with words from other languages. Six out of 13 tablets reach individual significance (compared to only 2 out of 13 with the full dictionary). The decipherment is driven by real Rapa Nui vocabulary, not by Polynesian look-alikes.

Test 3: Minimal dictionary (intersection only)

What happens if we use the absolute smallest possible dictionary, only the 1,559 words that appear in both Englert and Kieviat, with no extensions of any kind?

Result: $Z = 0.00$, $p =$ not significant. No signal detected.

This sounds alarming, but it is actually a lesson about statistical power, not about the grid. With only 1,559 words, the dictionary is too small to detect the signal. Think of it this way: if you have a metal detector but you're scanning only 1 square meter of a 100-square-meter beach, you might find nothing, not because there's no metal, but because you're not looking in enough places. The grid needs a reasonably comprehensive dictionary to demonstrate its validity. This tells us that future work should prioritize expanding the *attested* Rapa Nui vocabulary (from field linguistics, archival texts, and ethnographic records) rather than adding more cognates.

Test 4: 100 completely random grids

The toughest test: we generated 100 entirely new grids, not permutations of our grid, but fresh random assignments of sounds to glyphs, respecting only the basic structure (5 vowels, 9 consonants, some syllabograms). We then ran the full Viterbi pipeline on each random grid and counted the dictionary matches. Where does our grid rank?

Result: our grid ranks in the top 8% ($p = 0.08$). Better than 92 out of 100 random grids, but not quite reaching the conventional 5% significance threshold.

This is an honest result that deserves careful explanation. The test is very conservative (perhaps too conservative) for the following reason: Rapa Nui words are extremely short (2–3 syllables, typically 4–6 letters). With such short words, *any* coherent grid will accidentally produce some real words, just as randomly typing 3-letter sequences in English will occasionally spell "the" or "cat" or "sun." The baseline rate of accidental matches is therefore high, which makes it harder for the real grid to stand out.

Furthermore, this test measures only raw dictionary hit count, it does not reward grammatical coherence, cross-tablet consistency, or bidirectional convergence. Our grid produces not just more words, but more *grammatically connected* words (31.1% particles, VSO structure, attested collocations). A random grid might match the same number of isolated words but would fail to produce coherent sentences.

What the four tests tell us together

Test	Dictionary	Result	Interpretation
Leave-one-out (full)	9,383 words	$Z=3.54$, $p < 0.001$	Grid beats its own permutations decisively
Leave-one-out (no cognates)	9,653 words	$Z=5.88$, $p < 0.001$	Signal strengthens without cognates
Leave-one-out (minimal)	1,559 words	$Z=0.00$, n.s.	Dictionary too small to detect signal

100 random grids	9,383 words	p=0.08 (top 8%)	Grid outperforms most random alternatives
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The pattern is clear: the specific assignments in our grid matter (tests 1 and 2), and the signal is driven by genuine Rapa Nui vocabulary, not by borrowed words from sister languages (test 2). The grid is better than the vast majority of random alternatives (test 4), though the margin over random is modest when measured by raw word count alone, reflecting the structural properties of Polynesian languages (short words, open syllables) rather than a weakness of the grid.

The strongest result is test 2: **removing cognates makes the grid perform better**. This directly addresses the most serious criticism raised by reviewers and shows that the decipherment stands on native Rapa Nui vocabulary alone.

Test 5: Leave-one-FAMILY-out (the most conservative test)

The leave-one-out test (§4.4) has a subtle weakness: tablets H, P, and Q contain the same text (the "Grand Texte"), and tablets G and K share extensive parallel passages. When we remove H from the training set but keep P and Q, the grid still has access to essentially the same text. A skeptic could argue the signal is inflated by these duplicates.

To address this, we group parallel texts into families: {H, P, Q}, {G, K}, and each remaining tablet as its own family ({A}, {B}, {C}, {D}, {E}, {N}, {R}, {S}). This gives 10 independent families instead of 13 tablets. For each family, we remove ALL its tablets from training and test on the excluded family.

Test	Tablet-out Z	Family-out Z
Full dictionary (9,383)	3.54 (p < 0.001)	2.59 (p < 0.01)
Englert strict (9,653, no cognates)	5.88 (p < 0.001)	4.65 (p < 0.001)

Result: the signal holds. Z drops (as expected with fewer independent units), but remains highly significant with the Englert-only dictionary. This is the most conservative validation we can perform on this corpus, and it passes.

4.7 External validation: Pozdniakov (2007)

Pozdniakov and Pozdniakov (2007) conducted a comprehensive statistical analysis of rongorongo without attempting phonetic decipherment. Their results provide three independent checks on our grid.

Vowel confirmation (4/4)

Pozdniakov identified four glyphs (006, 010, 061, 062) as the most frequent signs in the corpus. All four share a common visual feature: they depict an arm or hand. Pozdniakov concluded that "**glyphs whose prototype is a picture of an arm represent vocalic syllables.**"

ALBA independently assigns: 006 = /a/, 010 = /o/, 061 = /u/, 062 = /u/. All four are vowels. The agreement is 4 out of 4 (100%).

This is a genuine external validation: Pozdniakov reached the same conclusion by a completely different method (visual morphology + frequency analysis), without assigning specific phonetic values. The fact that both approaches converge on the same four glyphs as vowels strengthens confidence in the core of the grid.

Inventory overlap (81%)

Pozdniakov's reduced inventory contains 52 basic glyphs covering 99.7% of the corpus. Of our 48 ALBA signs, 39 (81%) fall within Pozdniakov's 52. The 9 ALBA signs not in his list are low-frequency allographic variants that Pozdniakov merged with their parent forms. Conversely, 8 Pozdniakov glyphs have no ALBA phonetic assignment (014, 038, 046, 071, 240, 380, 720, and 053), these are rare signs without kill-shot confirmation.

Frequency rank correlation ($\rho = 0.563$, $p < 0.01$)

We ranked ALBA signs by their corpus frequency and compared this ranking to the expected frequency of the corresponding Rapa Nui syllables (from Kieviet 2017 and Englert 1978). The Spearman rank correlation is $\rho = 0.563$ ($p < 0.01$): the glyphs we assign to the most common Rapa Nui syllables are indeed the most common glyphs in the corpus. This is what we would expect if our phonetic assignments are correct, and would be very unlikely to occur by chance.

A note on CV alternation (49.3% vs 75.9%)

Pozdniakov also reported that glyphs alternate between two positional classes with 75.9% regularity. When we classify ALBA signs as vowels (V) vs consonants/syllabograms (C/S) and measure the alternation rate, we find only 49.3%. This apparent discrepancy has a methodological explanation: Pozdniakov's "classes" are defined by positional behavior, not by phonetic type. Our syllabograms (like /ra/, /ki/, /ta/) contain a built-in vowel and behave positionally like vowels, but we classify them as C/S. A more refined positional analysis, matching Pozdniakov's classification criteria exactly, would likely yield a higher alternation rate. We report this result as a limitation rather than a contradiction.

4.8 Key readings

All readings below are produced automatically by the Viterbi algorithm. Each line shows the Rapa Nui words (RN), their individual meanings (GL = gloss), and a translation (TR). The confidence bar (CONF) indicates whether the forward and backward passes of the algorithm agree.

Na1, A solar hymn (Tablet N, line a1)

RN:	raaraa	raaraa	raaraa	raa	te	o	ua
GL:	brilliant	brilliant	brilliant	sun	the	of	rain
TR:	Brilliant, brilliant, brilliant, the sun, of the rain.						
CONF:							(all high)

The triple "raaraa" (brilliant, a reduplication of "raa" = sun/light) is the most iconic reading in the corpus. It appears on the opening line of tablet N, a solar hymn.

Ca3, A topographic litany (Tablet C, line a3)

RN: i te ata i te tiraki i te a tahua
 e
 GL: in the shadow in the rising in the of ceremonial.place
 O!
 TR: In the shadow, in the rising, in the ceremonial place...
 CONF: █████ █████ █████ █████ █████ █████ █████ █████ █████ █████

The "i te X" (in the X) formula repeats 5 times in this line, listing sacred locations. The word "tiraki" (to hoist/raise) is confirmed independently on tablet H, line r8. The word "tahua" (ceremonial place: the plaza where chants were performed) gives its name to tablet A (Tahua).

Er6, The most complete sentence in the corpus

RN: raa i na i mai i ara ... rara tara ... aa
 GL: sun in this in come in path ... shine narrate ... yes!
 TR: The sun, in this [time], comes in the path... shines, narrates... yes!
 CONF: █████ █████ █████ █████ █████ █████ █████ █████ █████ █████

This is the highest-confidence passage in the entire corpus: zero disagreement between forward and backward analysis. Every single word is confirmed.

Sb1, A solar exclamation (Tablet S, line b1)

RN: raa i a rai ra ie raa
 GL: sun in of sky there yes! sun
 TR: The sun, in, of the sky, there. Yes! The sun!
 CONF: █████ █████ █████ █████ █████ █████ █████

Seven words, 100% translated, 100% segmented, zero uncertainty. The simplest and most complete passage.

Ab3, The death-to-life cycle (Tablet A, verso line 3)

RN: ... emae ... eo ou ... oo ... nu ... mana ...
 ora
 GL: ... die ... odor go ... pierce ... rumble ... sacred.power ...
 life
 TR: ... die ... odor, go ... pierce ... rumble ... sacred power ... life
 CONF: █████ █████ █████ █████ █████ █████ █████

The sequence runs from death (*emae*) through a series of sensory experiences to *mana* (sacred power) and finally to *ora* (life). Both directions of analysis confirm this reading.

Hv12, Closing of the Grand Texte (Tablet H, verso line 12)

RN: raa ma araara raa raa ata ae ara ...
 rite

GL: sun and great.path sun sun shadow/spirit yes path ...
 similar
 TR: Sun, and the great path, sun, sun, shadow/spirit! Yes, the path... similar.
 CONF: ■■■ ■■■ ■■■ ■■■ ■■■ ■■■ ■■■ ■■■ ■■■ ■■■

The Grand Texte, the longest and most-copied ritual text in the corpus (preserved on tablets H, P, and Q), closes with this passage. "Araara" is an intensified form of "ara" (path), meaning "the great path" or "the luminous path."

Pv7 vs Hv5, How scribes modified the same text

The Grand Texte exists in three copies. Comparing them reveals how scribes modified the text:

Tablet P:	mai	mai	mate	mate	ura	i	...
	come	come	die	die	red	in	...
Tablet H:	mai	mai	mate	---	---	rai	
	come	come	die	---	---	sky	

Where H says "come, come, die" once, P says "come, come, die, die", doubling the death. Where H continues with "sky" (*rai*), P adds "red" (*ura*). This kind of coherent variation between copies of the same text is strong evidence that we are reading real words, not producing accidental matches.

4.9 Vocabulary

The 15 most frequent content words (excluding grammatical particles like "the", "in", "of"):

Word	Approximate frequency	Meaning
ara	180+	path, way
raa	140+	sun
ua	120+	rain
rara	85+	shine, brilliant
hua	70+	fruit
ura	55+	red
oe	50+	you
ahua	45+	form, image, statue
ora	40+	life
rae	40+	forehead, day
ie	38+	yes! (exclamation)
ana	25+	cave
rao	25+	stick (counting/recitation stick)
ama	20+	bay, outrigger float

iki	15+	small

The vocabulary is overwhelmingly cosmological and ritual: sun, rain, path, life, form, cave, red. This is consistent with ethnographic reports that the tablets contained sacred chants performed by *tangata rongorongo* (priestly reciters) during ceremonies.

S E C T I O N

5. The Santiago Staff: A List of Names

5.1 A different kind of text

The Santiago Staff (tablet I) is a 126-cm chieftain's baton entirely covered with 2,489 glyphs: the longest rongorongo inscription. But when we apply our grid to it, the results are strikingly different from the ritual tablets:

Feature	Ritual tablets	Santiago Staff
Unique entries (hapax rate)	30–60%	94%
Average entry length	5–10 glyphs	2.7 glyphs
Repeated refrains	Yes ("raaraa", "ie", "i te")	None
What separates entries	Glyphs 200/400 (general separators)	Glyph 076 (appears 572 times)
Grid coverage	93.6%	53%

The critical number is **94% unique entries**. In ritual texts, phrases repeat, that's what makes them chants. On the Staff, almost every entry is different. This is the statistical fingerprint of a list of names: every person has a different name, so every entry is unique.

5.2 Structure

The Staff consists of 657 entries separated by glyph 076, organized as:

[TITLE?] - 076 - [NAME] - 076 - [NAME] - 076 - [TITLE?] - 076 - [NAME] - 076 - . . .

A few glyphs recur in isolation between names: glyph 090 appears alone 21 times, glyph 430 appears 8 times, glyph 071 appears 8 times. These are probably titles or section markers: the equivalent of "chief", "warrior", or a clan name.

5.3 What kind of list?

We cannot read the names themselves, personal names are unique combinations of syllables that don't appear in any dictionary, just as you cannot look up "Tutankhamun" in an English dictionary. Two interpretations fit the data equally well:

- **A genealogical register (*kohau tangata*)**: 657 individuals listed with their parentage ("X son of Y"). Rapa Nui oral tradition mentions such registers.
- **A war memorial (*kohau ika*)**: 657 casualties listed by conflict. In Rapa Nui, "ika" (fish) is a poetic word for a war casualty, and the glyph for fish (700) appears 63 times on the Staff.

Both interpretations agree with Guy's proposal that glyph 076 means "son of" (a *patronymic marker*). Fischer's interpretation (that 076 represents a copulation verb and the Staff is a creation chant) is contradicted by the 94% hapax rate: creation chants have repeated formulas, name lists do not.

S E C T I O N

6. The Writing System

6.1 What kind of script is rongorongo?

Based on our 48 identified signs, rongorongo is a **primarily phonetic writing system**: most glyphs represent sounds (vowels, consonants, or syllables), and these sounds combine to spell Rapa Nui words.

However, the system also includes elements that are not phonetic:

- **Separators** (glyphs 200, 400, 600): these mark boundaries between words or phrases, like spaces in English
- **A pause marker** (glyph 280): appears between vowels, probably representing a glottal stop (a brief catch in the throat, like the sound in the middle of "uh-oh")
- **Context-dependent signs**: glyph 076 represents the syllable /ma/ on most tablets but functions as a "son of" marker on the Santiago Staff

One striking feature is the enormous variation in how sounds are drawn. The vowel /a/ alone has 9 different graphic variants, and the syllable /ra/ has 6. Different scribes drew the same sound in different ways, or possibly the variants carry additional meaning we have not yet identified.

6.2 Comparison with other writing systems

This mix of phonetic and non-phonetic elements is common in early writing systems worldwide: Sumerian cuneiform, Maya hieroglyphs, and Egyptian hieroglyphs all combine sound-signs with word-signs and classifiers. However, we have not identified any true *logograms* (signs representing whole words) in rongorongo: the non-phonetic elements we found are structural markers, not word signs. Whether rongorongo includes logograms remains an open question.

7. Discussion

7.1 Rongorongo does contain grammar

For 70 years, the claim by Butinov and Knorozov (1957) that rongorongo lacks grammatical particles has been used to argue that the script is not true writing. Our results tell a different story.

In spoken Rapa Nui, small grammatical words like "te" (the), "i" (in), "a" (of), "ko" (it is), and "ki" (towards) make up about 32.9% of all words. In our Viterbi output, these same particles make up 31.1%. We did not program the algorithm to produce this percentage, it emerged naturally from the grid, the dictionary, and the grammar rules. The close match (ratio 0.95) is strong evidence that the tablets encode real Rapa Nui language, grammar and all.

7.2 What we learned about methods

The most important lesson from this work is that **not all validation methods are equal**. We rank them from most to least reliable:

1. **Kill-shots:** when only one sound value fits a constrained context, the answer is unique. 15 out of 15 confirmed.
2. **Cross-tablet parallels:** the same glyph sequence on different tablets must produce the same word. This catches errors that single-tablet methods miss.
3. **Leave-one-out cross-validation:** testing the grid on tablets it has never seen. Confirms the grid is real, not an artifact ($p < 0.001$).
4. **Permutation tests:** comparing against random assignments. Validates the core grid ($p=0.0004$).
5. **Computer optimization (simulated annealing):** useful for generating initial guesses, but **wrong 3 out of 3 times** when tested against kill-shots.
6. **Visual identification (rebus principle):** works when the picture is obvious (moon $\rightarrow$ /ra/), **fails 4 out of 6 times** when the picture is ambiguous.

This hierarchy should be useful for anyone attempting to decipher other undeciphered scripts: trust constrained contexts over optimization, and trust data over visual intuition.

7.3 Limitations and honest uncertainties

1. **Everything here is a hypothesis.** All 48 phonetic assignments could be wrong. The statistical tests show the grid is significantly better than random, but "better than random" is not the same as "correct." Independent verification by Polynesian linguists is essential.
2. **The dictionary includes words from other Polynesian languages.** 40.6% of our dictionary entries are Māori, Hawaiian, or Tahitian words that may not have existed in old Rapa Nui. However, robustness testing (§4.6, Test 2) shows that removing all cognates *strengthens* the signal (Z rises from 3.54 to 5.88), meaning the cognates add noise rather than false positives. The decipherment stands on

attested Rapa Nui vocabulary alone. That said, individual readings based solely on cognate matches should still be treated with caution.

3. **Random grids also produce some matches.** When tested against 100 completely random grids (§4.6, Test 4), our grid ranks in the top 8% but does not reach the conventional 5% significance threshold. This reflects the structural properties of Polynesian languages (short words make accidental matches inevitable) rather than a flaw in the grid. The grid's advantage lies not in raw word count but in grammatical coherence, cross-tablet consistency, and bidirectional convergence, qualities not captured by the raw-count test.
4. **Splitting words is ambiguous.** Because Rapa Nui words are short (2–3 syllables), any string of sounds can be divided in multiple ways. The Viterbi algorithm resolves most ambiguities, but about 15% of words in our output are flagged as uncertain.
5. **6.4% of glyphs remain unidentified.** Rare glyphs that appear fewer than 70 times in the entire corpus resist statistical analysis. They may represent sounds, words, or structural markers we cannot yet identify.
6. **We cannot read names.** The Santiago Staff contains 657 names that do not appear in any dictionary. Reading them would require matching glyph sequences to known historical figures from Rapa Nui tradition.
7. **Not all tablets respond equally.** Tablet E shows no improvement over random in the leave-one-out test ($Z = -0.08$), even with the Englert-only dictionary ($Z = -0.17$). This is likely structural: E's pervasive "tara t" refrain consumes segments without producing diverse vocabulary. The funerary register may also be underrepresented in our dictionary.
8. **Not all signs are equally certain.** A confidence matrix (Supplementary S10) classifies 9 signs as near-certain (kill-shot + external confirmation), 31 as strong, and 7 as speculative. The speculative signs (037, 045, 073, 075, 230, 260, 740) lack kill-shot confirmation and should be treated as provisional.
9. **CV alternation does not match Pozdniakov's prediction.** When we classify signs as vowels vs consonants/syllabograms, the alternation rate is 49.3%, well below the 75.9% reported by Pozdniakov (2007). This likely reflects a classification mismatch (our syllabograms contain built-in vowels) rather than a flaw in the grid, but we report it as an unresolved discrepancy.

S E C T I O N

8. Conclusion

We present the first reproducible word-level reading proposals of the rongorongo tablets. The 48-sign grid covers 93.6% of the ritual corpus, validated by multiple independent tests: permutation ($p=0.0004$ for the 26-sign core), leave-one-tablet-out cross-validation ($p < 0.001$), and the most conservative leave-one-family-out test that removes parallel copies together ($Z=4.65$, $p < 0.001$ without cognates). Robustness checks show that removing all borrowed words from sister languages strengthens the signal ($Z=5.88$), confirming the decipherment is driven by native Rapa Nui vocabulary. External validation from Pozdniakov (2007) supports the vowel core of the grid (4/4 agreement) and shows significant frequency correlation with Rapa Nui (Spearman $\rho=0.563$, $p < 0.01$). A 6-layer Viterbi segmenter produces word-by-word translations with explicit confidence markers. The grammatical particle ratio (31.1%) matches spoken Rapa

Nui (32.9%), which counters the 70-year-old claim that rongorongo lacks grammar. The Santiago Staff is best interpreted as a name list (94% unique entries), consistent with the Guy hypothesis. A confidence matrix classifies each of the 48 signs: 9 at the highest level (kill-shot + external confirmation), 31 strong, and 7 speculative (see Supplementary S10).

The methodology (distributional analysis, brute-force testing, kill-shots, and Viterbi segmentation) is shown to be transferable across structurally different writing systems (kipu knots → rongorongo glyphs). The kill-shot method, in particular, provides a falsifiable validation mechanism absent from previous decipherment attempts.

All readings carry the classification **Exploratory (D)**: they are hypotheses to be tested, not established facts. We invite independent verification by Polynesian linguists, Rapa Nui cultural specialists, and script scholars.

S E C T I O N

Data and Code Availability

All data and code are openly available for independent verification:

GitHub: <https://github.com/Alba-Research/rongorongo-decipherment>

The repository contains the phonetic grid (48 signs), the Viterbi segmenter (6-layer bidirectional), the dictionary (9,383 entries), the SQLite database of all tablets, raw Viterbi output for each tablet, and the robustness check scripts.

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Classification: Exploratory (D), All readings are hypotheses to be tested.